

Packages in Java Demonstration

Title: Packages – Defining, Creating, Accessing and Importing Packages in Java

Objective/Aim:

To define a package named "number", create a "Roman" class inside it, implement the "romanToInteger()" method, and import the method in another class to convert a Roman numeral into an integer.

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1]
- Minimum 4 GB RAM[cite: 1]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

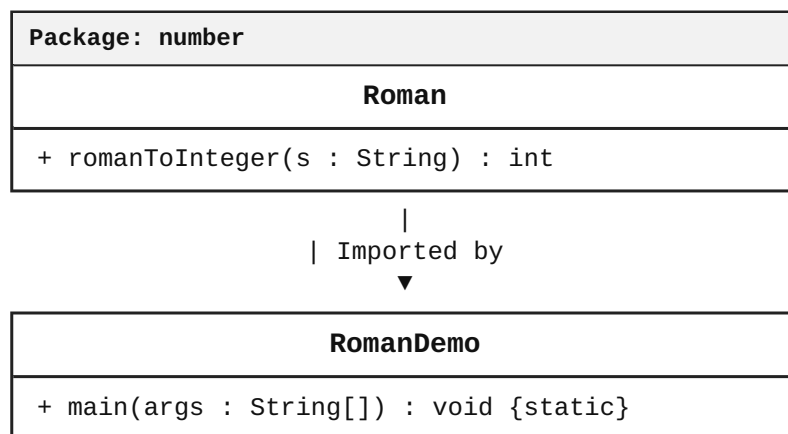
Software

- Operating System: Windows/Linux/macOS[cite: 1]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1]
- Visual Studio Code / Eclipse IDE or any Java IDE[cite: 1]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Start the program.
2. Define a package named `number`.
3. Create a public class named `Roman` inside the `number` package.
4. Implement the `romanToInteger(String s)` method to parse each Roman numeral and compute its decimal integer value.
5. Import the `number.Roman` class into another class named `RomanDemo`.
6. Store the Roman numeral "LVIII" in a `String` variable.
7. Create an instance of `Roman`, call `romanToInteger()`, and display the result.
8. Stop the program[cite: 2].

UML class Diagram



Relationship: RomanDemo imports and accesses the public class Roman located in the user-defined package number.

Source Code:

File 1: Roman.java

```
package number;

public class Roman {

    public int romanToInteger(String s) {
        int result = 0;
        int previous = 0;

        for (int i = s.length() - 1; i >= 0; i--) {
            int value = 0;

            switch (s.charAt(i)) {
                case 'I': value = 1; break;
                case 'V': value = 5; break;
                case 'X': value = 10; break;
                case 'L': value = 50; break;
                case 'C': value = 100; break;
                case 'D': value = 500; break;
                case 'M': value = 1000; break;
            }

            if (value < previous) {
                result -= value;
            } else {
                result += value;
            }

            previous = value;
        }

        return result;
    }
}
```

```
}  
}
```

File 2: RomanDemo.java

```
import number.Roman;  
  
public class RomanDemo {  
    public static void main(String[] args) {  
        Roman obj = new Roman();  
  
        String roman = "LVIII";  
        int result = obj.romanToInteger(roman);  
  
        System.out.println("Roman Numeral: " + roman);  
        System.out.println("Integer Value: " + result);  
    }  
}
```

Compilation & Execution Commands:

Save files as: number/Roman.java and RomanDemo.java

Compilation: javac number/Roman.java RomanDemo.java

Execution: java RomanDemo

Output Screenshot:

```
PS C:\Users\mmeen\RAVI> javac number/Roman.java RomanDemo.java  
PS C:\Users\mmeen\RAVI> java RomanDemo  
Roman Numeral: LVIII  
Integer Value: 58  
PS C:\Users\mmeen\RAVI>
```

Result:

Thus, the number package was created successfully, the Roman class was defined, and its `romanToInteger()` method was imported and accessed from another class to convert the Roman numeral "LVIII" into the integer "58".